

Chapter 7

INTRO to

MICROBIOLOGY



1. Rewrite the following statements using correct of the options and EXPLAIN the completed statements.

(gluconic acid, coagulation, amino acid, acetic acid, clostridium, lactobacilli)

- i. Process of coagulation of milk proteins occurs due to lactic acid.
- ii. Harmful bacteria like clostridium in the intestine are destroyed due to probiotics.
- iii. Chemically, vinegar is acetic acid.
- iv. Salts which can be used as supplement of calcium and iron are obtained from gluconic acid acid.

EXPLANATION

- i. The process of coagulation of milk proteins occurs due to lactic acid produced by Lactobacillus, which lowers the pH of milk, causing the proteins to aggregate into curds, separating them from the liquid whey.
- ii. Probiotics are good bacteria that fight harmful ones like Clostridium in our gut by releasing natural substances that kill them and by making it harder for them to grow
- iii. Vinegar is mainly composed of acetic acid, which gives it its sour taste and acidic properties. It is produced through the fermentation process of ethanol by acetic acid bacteria.

- iv. Salts such as calcium gluconate and iron gluconate can be used as supplements to provide calcium and iron, respectively. These salts are obtained by reacting gluconic acid with calcium and iron compounds.

2. Match the pairs

'A' group		'B' group
a. Xylitol		1. Pigment
b. Citric acid		2. To impart sweetness
c. Lycopene		3. Microbial restrictor
d. Nycin		4. Protein binding emulsifier
		5. To impart acidity

(a – 2), (b – 5), (c – 1), (d – 3)

3. Answer the following.

- a. Which fuels can be obtained by microbial processes? Why is it necessary to increase the use of such fuels?
- Various fuels like methane, ethanol, hydrogen gas, etc. are obtained by microbial processes.
 - Gaseous fuel- methane can be obtained by microbial anaerobic decomposition of urban agricultural and industrial waste.
 - Ethanol, an alcohol is a clean (smokeless) fuel obtained during fermentation of molasses by the yeast- *Saccharomyces*.

- iv. Hydrogen gas is considered to be the fuel of future. Hydrogen gas is released during bio-photolysis of water in which bacteria perform the photoreduction.
- v. It has become necessary to increase the use of such fuels because they are better for the environment, don't produce as much pollution, and can be made from various sources, including waste materials. This helps us reduce our reliance on fossil fuels that harm the planet.

b. How can the oil spills of rivers and oceans be cleaned?

- i. Spilling of petroleum oil occurs in ocean due to various reasons. This oil may prove fatal and toxic to aquatic organisms.
- ii. Bacteria like *Pseudomonas* spp. and *Alcanovorax borkumensis* have the ability to destroy the pyridines and other chemicals.
- iii. Hence, these bacteria are used to clear the oil spills. These are called as hydrocarbonoclastic bacteria (HCB).
- iv. HCB decompose the hydrocarbons and bring about the reaction of carbon with oxygen.
- v. CO_2 and water is formed in this process.



c. How can the soil polluted by acid rain be made fertile again?

- i. Sulphuric acid is present in the acid rain and in materials coming out of mines.
- ii. Erosion of bridges, metal statues, occurs due to acid rain.
- iii. Sulphuric acid is the source of energy for some species of bacteria like Acidophillium spp. and Acidobacillus ferroxidans.
- iv. Hence, these bacteria can control the soil pollution occurring due to acid rain, and the soil can be made fertile again.



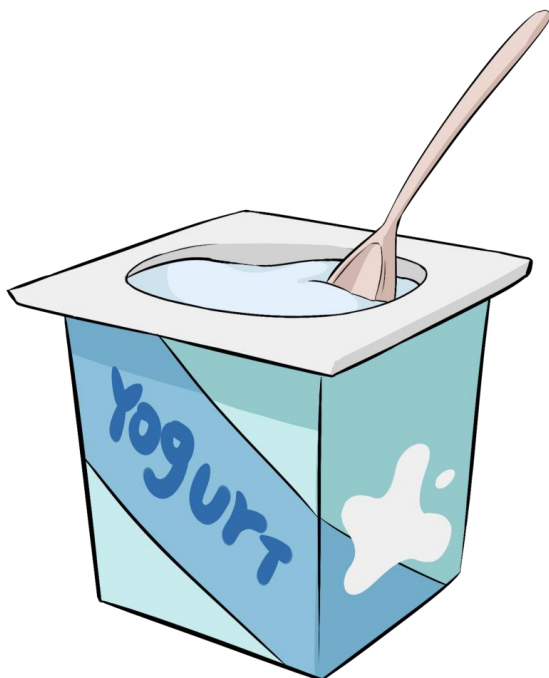
d. Explain the importance of biopesticides in organic farming.

- i. Fluoroacetamide-like chemicals are mixed with soil due to use of chemical pesticides in agriculture.
- ii. These prove to be harmful to other plants and animals as well as cause skin disease to humans. Therefore, biopesticides are used nowadays.
- iii. Bacterial and fungal toxins which can destroy pests and pathogens can be directly integrated into plants with the help of biotechnology.
- iv. Being toxic to insects, they do not consume the plants.
- v. Spinosad, a by-product of fermentation is a biopesticide.

e. Which are the reasons for increasing the popularity of probiotic products?

Following are the reasons for increasing the popularity of probiotic products:

- i. Probiotics are milk products containing active bacteria e.g. Lactobacillus, Acidophilus, Lactobacillus casei, Bifidobacterium bifidum, etc.
- ii. These microbes maintain the balance of intestinal microorganisms i.e. increase the population of microbes helping the digestion and decrease the population of harmful microbes (ex. Clostridium).
- iii. These products form the colonies of useful microbes in alimentary canal and control other microbes and their metabolic activities, improve resistance and lower the effects of harmful substances formed during metabolic activities.
- iv. Useful microbes become inactive due to antibiotics; probiotics make them active again.
- v. Nowadays, probiotics are used for treatment of diarrhoea and treatment of poultry also.



- f. How the bread and other products produced using baker's yeast are nutritious?**
- i. Compressed yeast is used in commercial bakery industry.
 - ii. It is available in dry, granular form for domestic use.
 - iii. Yeast produced for commercial use contains various useful contents like carbohydrates, fats, proteins, various vitamins and minerals.
 - iv. Due to this, bread and other products produced with the help of yeast become nutritive
- g. Which precautions are necessary for proper decomposition of domestic waste?**
- i. Domestic waste refers to the waste which is generated in our houses due to our daily activities.
 - ii. Domestic waste can be either biodegradable or non-biodegradable for example, vegetable and fruit peels are biodegradable whereas batteries, computers, etc. are non-biodegradable.
 - iii. Biodegradable products can be used for making composts which can be used in parks, plant pots etc.
 - iv. For non-biodegradable products, we need to practice the 3R's.
 - v. The 3R's are -
 - i. Reduce- Reduce the amount of waste produced.
 - ii. Reuse- Reusing a resource in a new way before throwing it.
 - iii. Recycle- Making something new from a material rather than throwing it away.

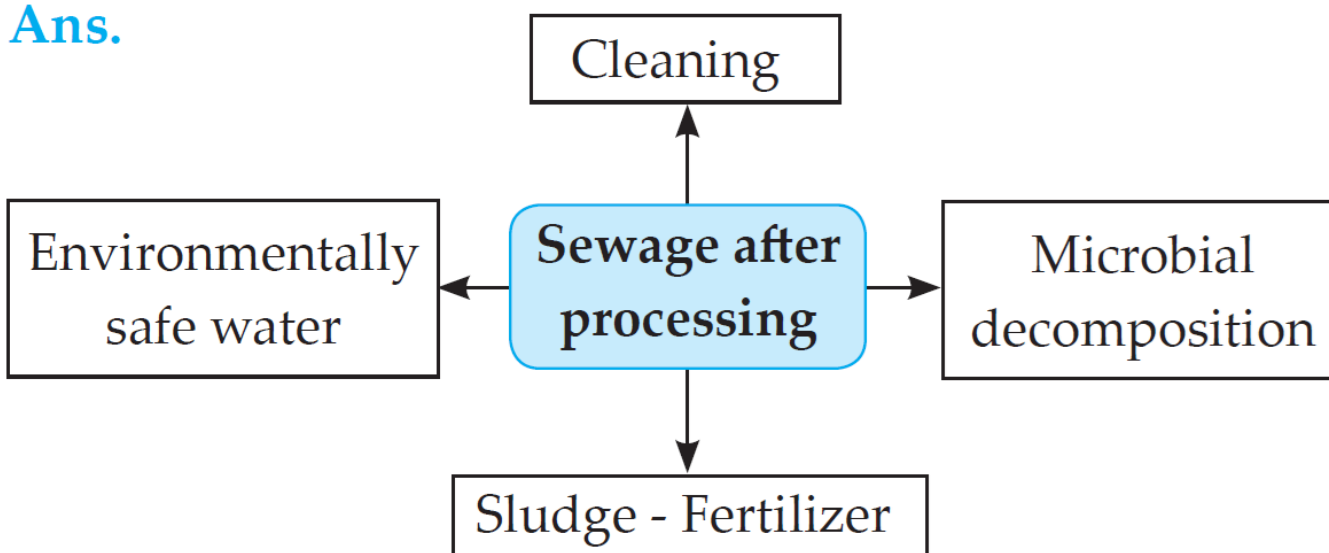
h. Why is it necessary to ban the use of plastic bags?

- i. Plastic bags are a major source of pollution, as they take hundreds of years to decompose and can end up in oceans and landfills, harming wildlife and ecosystems.
- ii. The production of plastic bags consumes valuable resources like oil and natural gas, which are non-renewable and contribute to climate change.
- iii. Plastic bags can leach harmful chemicals into food and water, posing health risks to humans and animals.
- iv. There are eco-friendly alternatives like reusable bags made from cloth or other biodegradable materials that can replace plastic bags, reducing waste and environmental harm.



4. Complete the following conceptual picture.

Ans.



5. Give scientific reasons.

a. Use of mutant strains has been increased in industrial microbiology.

- i. Mutant varieties of microorganisms are different from the original microorganism.
- ii. Mutant varieties are created by the insertion of genes of interest in a particular organism through a technique called genetic engineering.
- iii. Mutant microorganisms have characteristics that are not present in the original strain and thus make them useful for various purposes.
- iv. During industrial production of milk products, strict cleanliness and sterilization is essential as bacteria can be attacked by viruses. Hence, virus-resistant mutant varieties of bacteria are developed.

b. Enzymes obtained by microbial process are mixed with detergents.

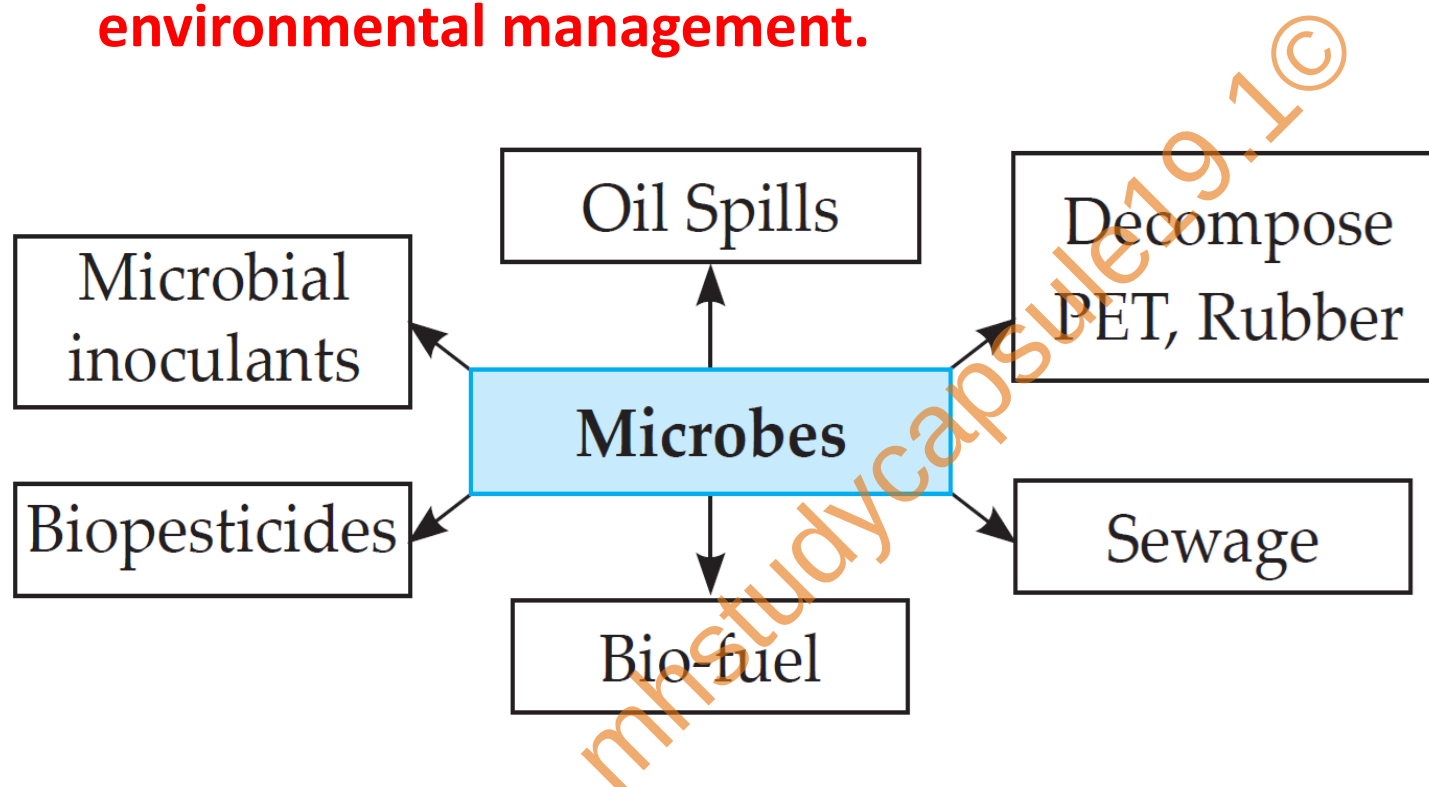
- i. Enzymes obtained by microbial process are active at low temperature, pH and pressure.
- ii. Hence enzymes mixed with detergents help in process of dirt/muck removal even at low temperature.

c. Microbial enzymes are used instead of chemical catalysts in chemical industry.

- i. Microbial enzymes are active at low temperature, pH and pressure, due to which energy is saved and erosion - proof instruments are also not necessary.

- ii. Enzymes carry out specific processes, hence unnecessary by-products are not formed due which expenses on purification are minimised.
- iii. Elimination and decomposition of waste materials is avoided and enzymes can be reused. Such enzymes are eco-friendly.
- iv. Therefore, microbial enzymes are used instead of chemical catalysts in chemical industry.

7. Complete the following conceptual picture related to environmental management.



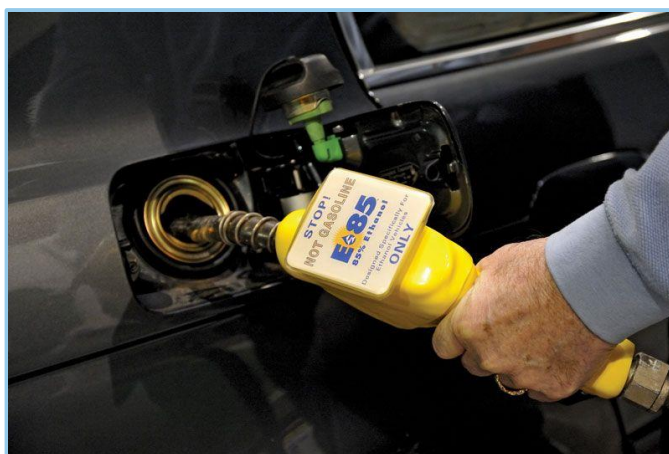
8. Answer the following.

a. What is role of microbes in compost production?

- i. Microbes play an important role in various aspects of our lives. One such important role is the formation of compost.
- ii. The biodegradable waste accumulated in urban areas is placed in landfill sites.
- iii. Landfill sites are created by digging large pits away from residential areas and the biodegradable waste is dumped into it.
- iv. Microbes present in soil and other top layers decompose the waste, especially aerobic bacteria and some kind of fungi.
- v. Best quality compost is formed after few days.
- vi. Such land filling sites can be reused after removal of compost.

b. What are the benefits of mixing ethanol with petrol and diesel?

- i. Ethanol an alcohol is a clean smokeless fuel. It is obtained during fermentation of molasses by yeast *Saccharomyces*.
- ii. Whenever ethanol is mixed with petrol and diesel, CO_2 emission from automobiles reduces by 30%. As a result, pollution decreases.
- iii. Since ethanol is a renewable, domestically produced transportation fuel, it reduces dependence on fossil fuels.



c. Which plants are cultivated to obtain the fuel?

Fuels can be obtained from cultivating plants like corn, sugarcane, wheat, palm oil, jatropha, soybean, beat, switchgrass and rapeseed.

d. Which fuels are obtained from biomass?

Fuels obtained from biomass:

- i. Solid fuels: Coal, dung, crop residue.
- ii. Liquid fuels: Vegetable oils, alcohol.
- iii. Gaseous fuels: Gobar gas, coal gas.

e. How does the bread become spongy?

- i. Different types of breads are produced from flour of cereals.
- ii. Dough is formed by mixing of baker's yeast – *Saccharomyces cerevisiae*, water, salt and other necessary materials with flour.
- iii. Due to fermentation of carbohydrates by yeast, sugar is converted in to carbon dioxide (CO_2) and ethanol ($\text{C}_2\text{H}_5\text{OH}$).
- iv. Dough rises up due to CO_2 and the bread becomes spongy.



INTEXT QUESTIONS

1. Which different microbes are useful to us?

- i. Lactobacilli is used for Yoghurt production.
- ii. Rhizobia living in root nodules of leguminous plants produce nitrogenous compounds from atmospheric nitrogen
- iii. Yeast - *Saccharomyces cerevisiae* is used for ethanol and bread production.

2. Which different products can be produced with the help of microbes?

- i. Products produced with the help of microbes are bread, cheese, wine, yoghurt, butter, probiotics and vinegar.
- ii. They are also used for producing chemicals, enzymes, nutrients, medicines like antibiotics and also for Bio-fuel.
- iii. They are used in garbage management and pollution control.

3. We use the fermentation process while conversion of milk into yoghurt. Which microbes are useful for this process?

- i. Milk is fermented with the help of Lactobacilli.
- ii. The lactobacilli convert lactose, the sugar in the milk, into lactic acid.
This process is called fermentation.

4. Which different types of cheese are used in western food like pizza, burger, sandwich, etc?

Cottage cheese, cream cheese, mozzarella cheese, cheddar cheese, parmesan cheese, etc. are used in western foods.

5. What is difference between those types of cheese?

- i. Cottage cheese, cream cheese, mozzarella cheese are freshly prepared soft cheese.
- ii. On storing for 3 to 12 months, semi-hard cheddar cheese is formed whereas after ripening for 12 – 18 months, very hard cheese called parmesan cheese is formed.

6. What for probiotic food is famous?

Probiotic food is famous for maintaining the balance of intestinal microorganisms, i.e. increasing the population of microbes helping in digestion and decreasing the population of harmful microbes like Clostridium.

7. In the earlier class, you had prepared the solution of dry yeast for observation of yeast. Which substance is prepared by its use on commercial basis?

The commercial production of bread and other bakery products need yeast. In wine and beer making also solution of yeast is required.

7. Which functions are performed by enzymes secreted in human digestive system? Give names of some such enzymes.

- i. Enzymes secreted in the digestive system perform the function of digestion.
- ii. Enzyme protease digests proteins. Enzyme amylase digests carbohydrates. Enzyme lipase digests fats.

8. Food materials like cold drinks, ice creams, cakes, juices are available in various colours and flavours. Whether these colours and flavours are really derived from fruits?

- i. These food materials do contain colours and flavours obtained directly from fruits.
- ii. However, most of the food products purchased from markets use colours and flavours which are artificially prepared.

9. What do you mean by antibiotic? Which precautions should be taken about their consumption?

Carbon compounds obtained from some bacteria and fungi for destroying or preventing the growth of harmful micro-organisms are called antibiotics.

Precautions:

- i. Antibiotics should be taken only when prescribed by a doctor.
- ii. Don't purchase any antibiotic from medical stores without a prescription from a doctor.

- iii. Don't consume antibiotics on your own to treat common diseases like a throat infection, common cold or influenza.
- iv. Even if you feel well before completion of the prescribed course of the antibiotic, you must continue and complete it.
- v. Don't suggest to others the antibiotics which were useful to you.

10. Which different materials are decomposed in biogas plant?

Agricultural waste, manure, municipal waste, plant material, sewage, green waste or food waste are decomposed in a bio-gas plant.

Which useful materials are obtained through it? Which is the fuel out of those?

Materials obtained are methane (CH_4), carbon dioxide (CO_2), small amount of hydrogen sulphide (H_2S), moisture. Slurry is also obtained from the biogas plant. Methane is the fuel out of those.

Decomposition occurs through which organisms?

Decomposition occurs through bacteria and fungi.

11. Why is it asked to segregate wet and dry waste in each home?

Waste must be segregated as dry and wet waste at homes because dry waste can be given to recycling units and wet waste can be converted into compost.

What is done with the segregated waste?

Dry waste is recycled, wet waste is sent to landfills for composting.

Which is most appropriate method of disposal of dry waste?

The most appropriate method of disposal of dry waste is recycling. Segregate the dry waste itself. Separate the commodities. Commodities like glass, metals, plastic can be recycled.

12. Which materials should not be present in garbage for its proper microbial decomposition?

- i. If there are non-biodegradable materials in the garbage , they will not decompose.
- ii. The plastic, glass, metals etc. will not undergo microbial decomposition, therefore, such items should not be there in the garbage.
- iii. The toxic matter, hazardous chemicals and e-waste should also be removed. If such materials are present in the garbage, the microbes will be killed and the entire process of decomposition will be affected.

13. How the sewage generated in your house or apartment is disposed off ?

- i. The sewage generated in our house is carried by the drainage pipes to municipal sewage treatment plants.
- ii. Here, primary, secondary and tertiary treatment is done on the sewage.
- iii. The safe water is then released into water bodies.



14. You must have seen or read the news of dead fishes or oily water accumulating at the seacoasts. Why does this happen?

- i. This happens due to oil spills. Oil spills can occur due to accidents of oil tankers that transport crude oil, spillage through offshore oil wells, etc.
- ii. The oil forms a film on the surface of water. It causes depletion of oxygen in the water.
- iii. It also contains hazardous hydrocarbons which results in the death of fish.



15. How the bacteria present in soil and root nodules of leguminous plants are useful?

Rhizobium Bacteria present in soil and root nodules of leguminous plants fix atmospheric nitrogen and convert into nitrites and nitrates and make it available to plants.

Previous Year Questions

1. Microbial enzymes are used instead of chemical catalyst in a chemical industry - **Exercise Question 5 - c**

2. Complete the following table:

Aspergillus niger

Molasses, corn steep liquor

Lactic acid

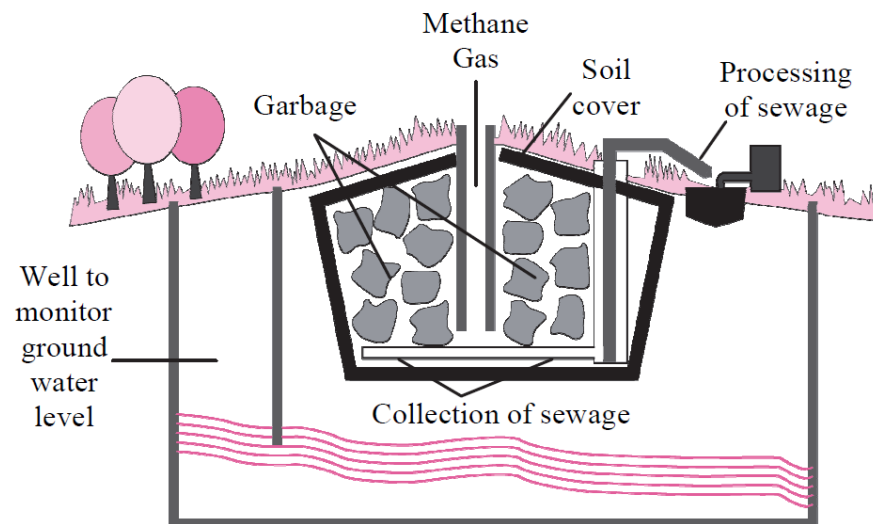
Itaconic acid

3. Complete the following conceptual picture: **Exercise Question 7**

4. Explain the term Bioinsecticides.

- i. Bioinsecticides are a type of pesticide that are derived from natural sources, such as plants, bacteria, fungi, or other organisms.
- ii. Bacterial and fungal toxins which can destroy pests and pathogens can be directly integrated into plants with the help of biotechnology.
- iii. Insects do not consume such plants, as they are toxic to insects.
- iv. Similar to bacteria, some species of fungi and viruses are useful as pesticides.
- v. Spinosad, a by-product of fermentation is a biopesticide.

5. Observe the following figure and answer the questions:



- a. Identify the process shown in the figure.
- b. Explain the process in short.

a. Process shown in the figure is of Modern landfill site.

b. Explanation:

- i. Degradable waste being accumulated in urban areas is used for this purpose.
- ii. Large pits are dug in open spaces far away from the residential area and those pits are lined with plastic sheets as a precaution against pollution of soil due to leaching of toxic and harmful materials.
- iii. Compressed waste is dumped in the pit. It is covered with layers of soil, saw dust, leafy waste and specific biochemicals.
- iv. Bioreactors are mixed at some places.
- v. Microbes present in soil and other top layers decompose the waste.
- vi. Completely filled pit is sealed with soil slurry.
- vii. Best quality compost is formed after few days.
- viii. Such land filling sites can be reused after removal of compost.

6. Which precautions are necessary for proper decomposition of domestic waste? - Exercise Question 3 - g

7. Complete the paragraph using proper words:

(Mechanical, Rhizobium, Aquatic, Toxic, CO₂, Nitrogen, Pseudomonas, Amoeba, Bacteria, Hydrocarbons)

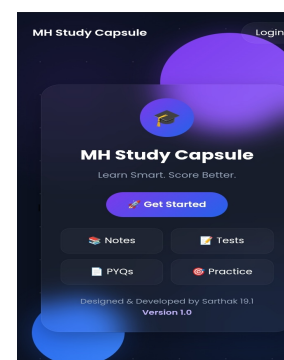
Spilling of petroleum oil occurs in ocean due to various reasons. This oil may prove fatal and toxic to aquatic organisms. It is not easy to remove the oil layer from surface of water by mechanical method. However, bacteria like Pseudomonas spp. and *Alcanivorax borkumensis* have the ability to destroy the pyridines and other chemicals. Hence, these bacteria are used to clear the oil spills.

8. Complete the following chart:

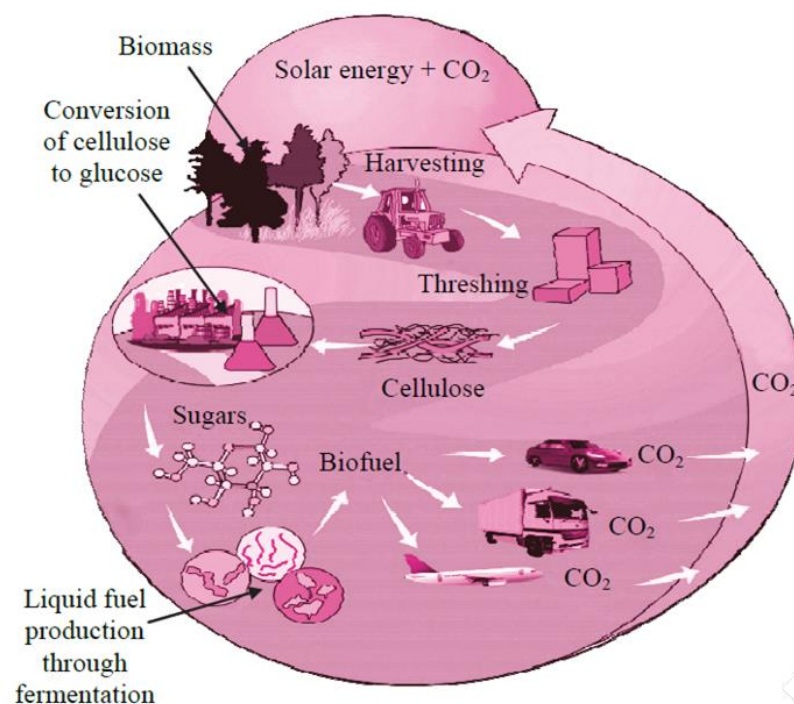
To impart acidity
Antioxidants, vitamins
Edible colours
Emulsifiers
Essence
Artificial sweetener (low calorie)

9. Why is it necessary to ban the use of plastic bags?

Exercise Question 3 - h



10. Answer the following:



- (1) Which process is shown in the above given picture?
- (2) Give *two* examples for liquid fuels and solid fuels obtained by this process.

i. Process shown in the figure is of Biofuel production.

ii. Examples:

Solid fuels (coal, dung, crop residue)

liquid fuels (vegetable oils, alcohol)

11. Answer the following:

a) Which substance is used to produce cheese?

b) Which enzyme was used earlier for cheese production?

c) Which enzyme is used to produce vegetarian cheese?

a) Abundantly available cow milk.

b) Rennet

c) Protease

12. What are the benefits of mixing ethanol with petrol and diesel?

Exercise Question 8 - b

13. Which fuels are obtained from biomass? Exercise Question 8 - d

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